MAJOR PROJECT 1

Design and Simulation of an 8-bit Arithmetic Logic Unit (ALU)

Project objective

To design and simulate an 8-bit ALU in Verilog that performs addition, subtraction, AND, OR and XOR.

Student Name: _____

Class / Section: _____

Roll Number: _____

1. Introduction

An Arithmetic Logic Unit (ALU) is a fundamental digital component in a processor. It performs arithmetic and bitwise logical operations on binary data. This project implements a simple 8-bit ALU using Verilog HDL and verifies it through simulation.

2. Objectives

• Design an 8-bit ALU with two 8-bit inputs and a 3-bit operation selector.
• Implement addition, subtraction, AND, OR and XOR.
• Generate CarryOut and Zero status flags.
• Verify results using a Verilog testbench and waveform viewer.

3. Inputs and Outputs

Signal	Width	Description
A	8 bits	First operand
B	8 bits	Second operand
Sel	3 bits	Selects operation
Result	8 bits	Operation result
CarryOut	1 bit	9th bit of arithmetic result; zero for logical operations
Zero	1 bit	1 when Result equals zero

4. Operation Selection Table

Sel	Operation	Expression
000	Addition	A + B
001	Subtraction	A – B
010	AND	A & B
011	OR	A | B
100	XOR	A ^ B

5. Design Methodology

A combinational always block and case statement select the operation. Addition uses a 9-bit temporary value to capture the ninth carry bit. Subtraction uses direct 9-bit unsigned subtraction, matching the convention in the project reference. Logical operations set CarryOut to zero. Zero is asserted whenever Result equals zero.

6. Tools Required

• Icarus Verilog: compiles and runs the simulation.
• GTKWave: displays the generated VCD waveform.
• A text editor: used to create and edit the .v files.

7. Simulation Procedure

1. Save alu.v and tb_alu.v in the same folder.
2. Open a terminal in that folder.
3. Compile: iverilog -o alu_tb alu.v tb_alu.v
4. Run: vvp alu_tb
5. View waveforms: gtkwave alu.vcd
6. Confirm the testbench prints PASS for each test.

8. Test Cases and Expected Results

Test	Inputs / Operation	Expected Result	Flags
Addition	15 + 10	25 (00011001)	CarryOut=0, Zero=0

Test	Inputs / Operation	Expected Result	Flags
Subtraction	20 – 5	15 (00001111)	CarryOut=0, Zero=0
AND	CC & AA	88 (10001000)	CarryOut=0, Zero=0
OR	CC | AA	EE (11101110)	CarryOut=0, Zero=0
XOR	CC ^ AA	66 (01100110)	CarryOut=0, Zero=0
Zero check	0 + 0	0 (00000000)	CarryOut=0, Zero=1
Carry check	250 + 10	4 (00000100)	CarryOut=1, Zero=0

9. Source Code

The complete synthesizable module is provided in alu.v and the automated tests are provided in tb_alu.v in the accompanying bundle.

10. Result and Conclusion

The 8-bit ALU supports the five required operations. The testbench verifies arithmetic and logical results, addition carry, and the Zero flag. Successful simulation demonstrates the intended behavior for the selected test cases.

11. Future Improvements

Possible extensions include NOT, NAND, shift operations, signed overflow detection, and FPGA implementation.